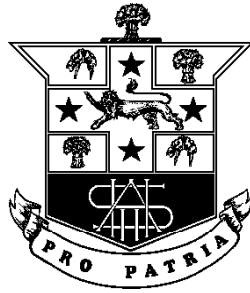


HURLSTONE AGRICULTURAL HIGH SCHOOL



MATHEMATICS – EXTENSION TWO

2011 HSC

ASSESSMENT TASK 3

Examiners ~ G Huxley, G Rawson

GENERAL INSTRUCTIONS

- | | |
|---|--|
| <ul style="list-style-type: none">• Reading Time – 3 minutes.• Working Time – 40 minutes.• Attempt all questions.• All necessary working should be shown in every question.• This paper contains two (2) questions. | <ul style="list-style-type: none">• Marks may not be awarded for careless or badly arranged work.• Board approved calculators may be used.• Each question is to be started in a new booklet.• This examination paper must NOT be removed from the examination room. |
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STUDENT NAME: _____

TEACHER: _____

QUESTION ONE 15 marks Start a SEPARATE booklet.

Marks

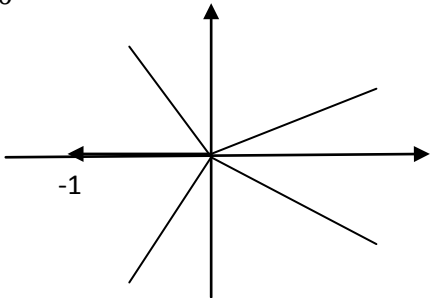
- (a) (i) Show that $P(x) = x^5 - 3x^4 + 4x^3 - 4x^2 + 3x - 1$ has $x = 1$ as a zero of multiplicity 3. **2**
- (ii) Verify that $x = i$ is also a zero of $P(x)$. **1**
- (iii) Hence solve the equation $P(x) = 0$. **2**
- (b) Let α, β, γ be the roots of the equation $x^3 - 2x^2 - 5x - 1 = 0$. Form an equation whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ **2**
- (c) (i) Solve $z^5 + 1 = 0$ by De Moivre's Theorem, leaving your solutions in modulus argument form. **2**
- (ii) Prove that the solutions of $z^4 - z^3 + z^2 - z + 1 = 0$ are the non-real Solutions of $z^5 + 1 = 0$. **1**
- (iii) Given that $z^n + \frac{1}{z^n} = 2n \cos \theta$, show that if $z^4 - z^3 + z^2 - z + 1 = 0$ then $4\cos^2 \theta - 2\cos \theta - 1 = 0$. **3**
- (iv) Hence find the exact value, fully simplified, of $\sec \frac{3\pi}{5}$. **2**

QUESTION TWO 15 marks Start a SEPARATE booklet.

Marks

- (a) Evaluate $\int_0^1 x e^{-x^2} dx$. 2
- (b) Using the substitution $u = e^x$, or otherwise,
find $\int \frac{e^x dx}{\sqrt{1 - e^{2x}}}$. 2
- (c) (i) Express $\frac{x^2 + 2x}{(x^2 + 4)(x - 2)}$ as the sum of partial fractions 2
- (ii) Hence, find $\int \frac{x^2 + 2x}{(x^2 + 4)(x - 2)} dx$ 2
- (d) Find $\int \frac{x^2}{1 + 4x^2} dx$ 3
- (e) Use the substitution $u = \sqrt{x - 1}$ to evaluate $\int_2^3 \frac{1 + x}{\sqrt{x - 1}} dx$ 4

Standard Integrals Sheet.

Year 12		Mathematics Extension 2	HSC Task 3 Jun 2011
Question No. 1 Solutions and Marking Guidelines			
Outcomes Addressed in this Question			
E4-uses efficient techniques for the algebraic manipulation required in dealing with polynomials.			
Outcome	Solutions	Marking Guidelines	
E4 (a)	(i) $P(x) = x^5 - 3x^4 + 4x^3 - 4x^2 + 3x - 1$ $P(1) = 1 - 3 + 4 - 4 + 3 - 1 = 0$ $P'(x) = 5x^4 - 12x^3 + 12x^2 - 8x + 3$ $P'(1) = 5 - 12 + 12 - 8 + 3 = 0$ $P''(x) = 20x^3 - 36x^2 + 24x - 8$ $P''(1) = 20 - 36 + 24 - 8 = 0$ $\therefore P(1) = P'(1) = P''(1) = 0$ So $x=1$ is a zero of multiplicity 3.	2 marks: Showing information given. Note: To give the absolutely perfect answer, you should also show that $P'''(1) \neq 0$, so that $x=1$ is not of multiplicity 4. 1 mark: Not showing that all 3 equations $=0$, Or not showing the substitution for this particular polynomial.	
	(ii) $P(i) = i^5 - 3i^4 + 4i^3 - 4i^2 + 3i - 1$ $= i - 3 - 4i + 4 + 3i - 1$ $= 0 + 0i$ Therefore i is a zero	1 mark: Correct answer	
	(iii) Coefficients are real, therefore complex roots are complex conjugate pairs. $P(x) = 0 \rightarrow x = 1, \pm i$	2 marks: Correct answer. 1 mark: Partially correct.	
	(b) Replace roots with their reciprocals, and simplify: $\left(\frac{1}{y}\right)^3 - 2\left(\frac{1}{y}\right)^2 - 5\left(\frac{1}{x}\right) - 1 = 0$ $1 - 2y - 5y^2 - y^3 = 0 \rightarrow x^3 + 5x^2 + 2x - 1 = 0$	2 marks: Correct answer. 1 mark: Partially correct.	
	(c) (i) $z^5 + 1 = 0$  $z = \text{cis} \pm \frac{\pi}{5}, \text{cis} \pm \frac{3\pi}{5}, \text{cis} \pi (\text{or } -1)$	2 marks: Correct answer. 1 mark: Partially correct: possibly zeroes missing of having incorrect arguments that still differ by $\frac{2\pi}{5}$.	

	(ii) $z^5 + 1 = (z + 1)(z^4 - z^3 + z^2 - z + 1)$ $z = -1$ is the solution of $z + 1 = 0$ Since $z = -1$ is the only real solution, the unreal solutions of $z^5 + 1 = 0$ are the solutions of $z^4 - z^3 + z^2 - z + 1 = 0$	1 mark: Correct solution, including the factorisation.
	(iii) Divide the equation by z^2. $z^2 \left(z^2 - z + 1 - \frac{1}{z} + \frac{1}{z^2} \right) = 0, \text{ where } z \neq 0$ $\therefore \left(z^2 + \frac{1}{z^2} \right) - \left(z + \frac{1}{z} \right) + 1 = 0$ $2 \cos 2\theta - 2 \cos \theta + 1 = 0$ $2(2 \cos^2 \theta - 1) - 2 \cos \theta + 1 = 0$ $4 \cos^2 \theta - 2 \cos \theta - 1 = 0$	3 marks: Correct solution. 2 marks: 1 error in working. 1 mark: Only one part of solution correct.
	(iv) $\cos \theta = \frac{1 \pm \sqrt{5}}{4}$ $\sec \frac{3\pi}{5} < 0, \quad \therefore \sec \frac{3\pi}{5} = \frac{4}{1 - \sqrt{5}} = -(1 + \sqrt{5})$	2 marks: Correct solution. 1 mark: Correct value of $\cos \theta$, or simplified calculation of $\sec \theta$ from incorrect $\cos \theta$.

Year 12		Mathematics Extension 2	HSC Task 3 Jun 2011
Question No. 2 Solutions and Marking Guidelines			
Outcomes Addressed in this Question			
E8- Applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae.			
Outcome	Solutions	Marking Guidelines	
E8	(a) $\int_0^1 x e^{-x^2} dx = -\frac{1}{2} \int_0^1 -2x e^{-x^2} dx$ $= -\frac{1}{2} \left[e^{-x^2} \right]_0^1$ $= \frac{e-1}{2e}$	2 marks: Correct answer, or equivalent 1 mark: Correct primitive, or correct calculation from incorrect primitive.	
	(b) $\int \frac{e^x dx}{\sqrt{1-e^{2x}}} = \int \frac{du}{\sqrt{1-u^2}}$ Let $u = e^x$ $du = e^x dx$ $= \sin^{-1} u + c$ $= \sin^{-1}(e^x) + c$	2 marks: Correct answer. 1 mark: Correct substitution, or correct calculation from incorrect substitution.	
	(c) (i) $\frac{x^2 + 2x}{(x^2 + 4)(x - 2)} = \frac{A}{x - 2} + \frac{Bx + C}{x^2 + 4}$ $x^2 + 2x \equiv A(x^2 + 4) + (Bx + C)(x - 2)$ $\rightarrow A = 1; \quad B = 0; \quad C = 2$ $\therefore \frac{x^2 + 2x}{(x^2 + 4)(x - 2)} = \frac{1}{x - 2} + \frac{2}{x^2 + 4}$	2 marks: Correct solution. Must include $B=0$ 1 mark: Partially correct.	
	(ii) $\int \frac{1}{x-2} + \frac{2}{x^2+4} dx = \ln x-2 + \tan^{-1} \frac{x}{2} + c$	2 marks: Correct answer, CFPA. 1 mark: One term correctly integrated.	
	(d) $\int \frac{x^2}{4x^2 + 1} dx = \int \frac{1}{4} \left(\frac{4x^2 + 1 - 1}{4x^2 + 1} \right) dx$ $= \frac{1}{4} \int 1 - \frac{1}{4x^2 + 1} dx$ $= \frac{1}{4} \left(x - \frac{1}{2} \tan^{-1} 2x \right) + c$	3 marks: Correct answer. 2 marks: Significant progress. 1 mark: Correct manipulation of fraction; or correct integration of one term.	

(e)	Let $u = \sqrt{x-1}$ $\therefore x = u^2 + 1$ $dx = 2u \, du$ $x = 3 \rightarrow u = \sqrt{2}$ $x = 2 \rightarrow u = 1$ $\int_2^3 \frac{1+x}{\sqrt{x-1}} dx = \int_1^{\sqrt{2}} \frac{2+u^2}{u} 2u \, du$ $= 2 \int_1^{\sqrt{2}} 2+u^2 \, du$ $= 2 \left[2u + \frac{u^3}{3} \right]_1^{\sqrt{2}}$ $= \frac{2}{3} (8\sqrt{2} - 7)$	4 marks: Correct solution, or equivalent. Includes: substitution; adjusting boundaries; finding primitive; calculating integral. 3 marks: Significant progress. 2 marks: 2 parts of solution correct. 1 mark: Only one part of solution correct..
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